

Fast Information Channel (FIC):

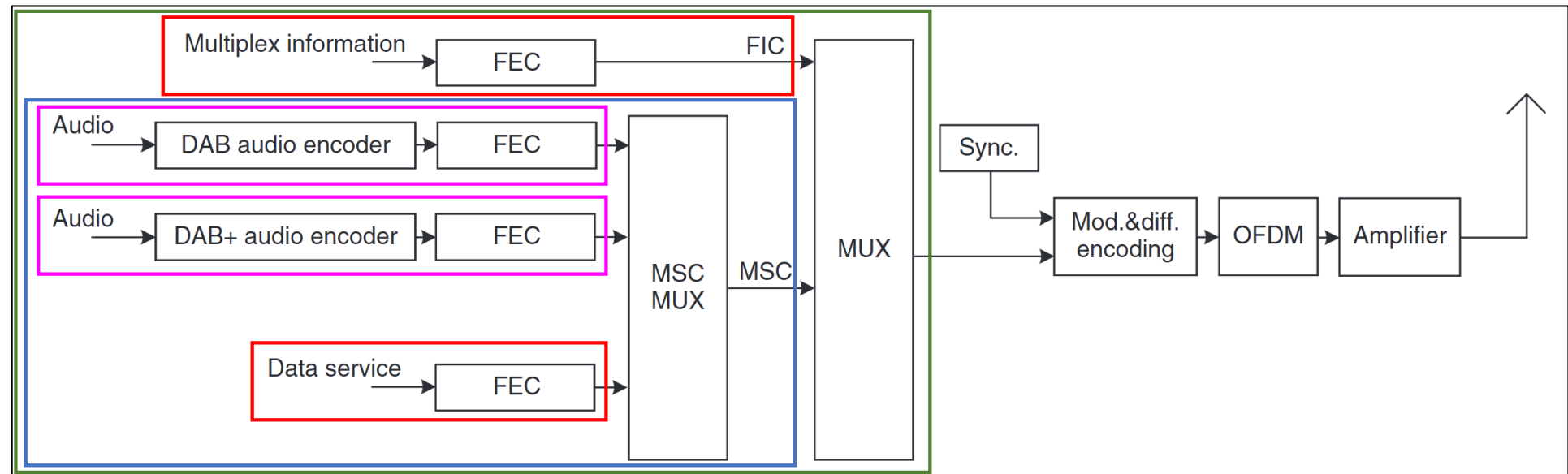
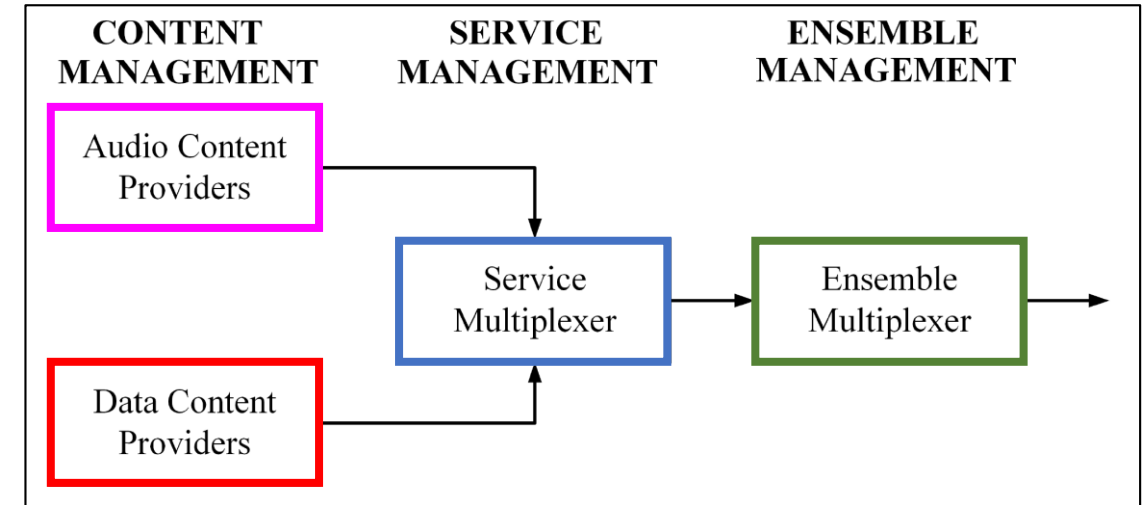
- Rapid access of information by a receiver (used to send the Multiplex Configuration Information (MCI) and Service Information)
- Transmitted at the beginning of each DAB frame using robust Forward Error Correction (**FEC**) parameters.

Main Service Channel (MSC):

- Carries the payload data (audio and data service components).
- Divided into a number of subchannels that contain the different services.
- Each sub-channel is then individually FEC encoded and has its own time interleaver,

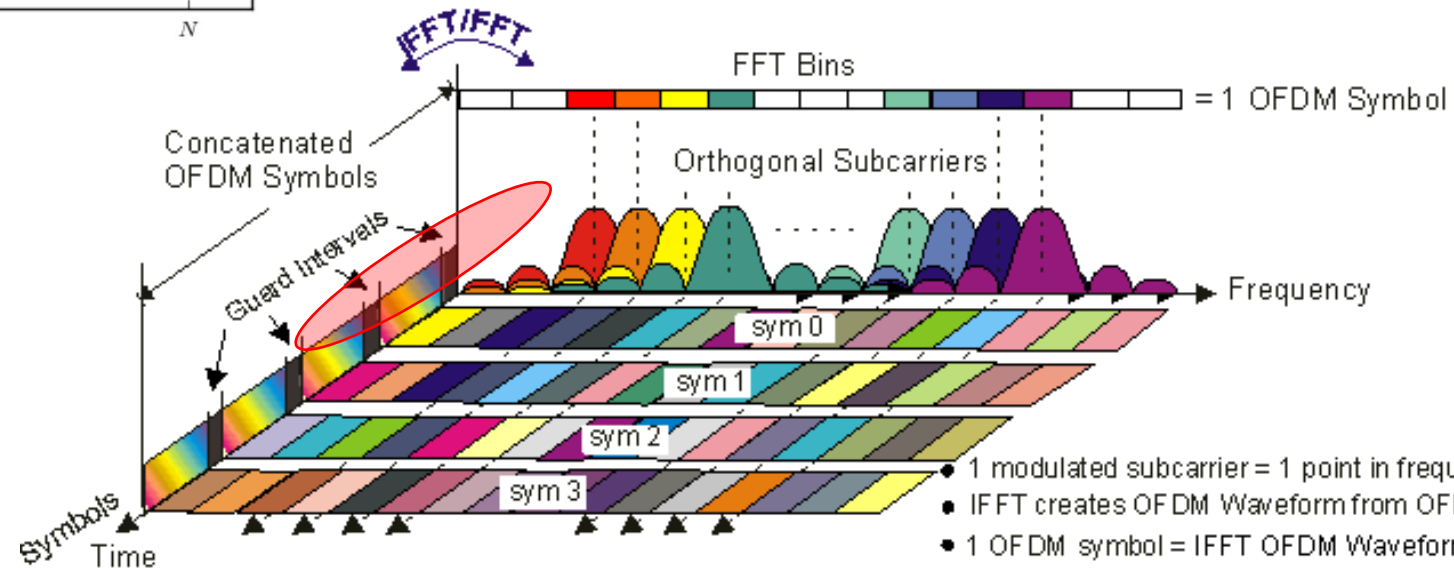
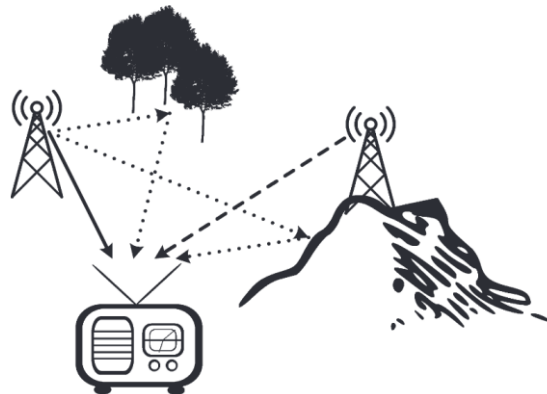
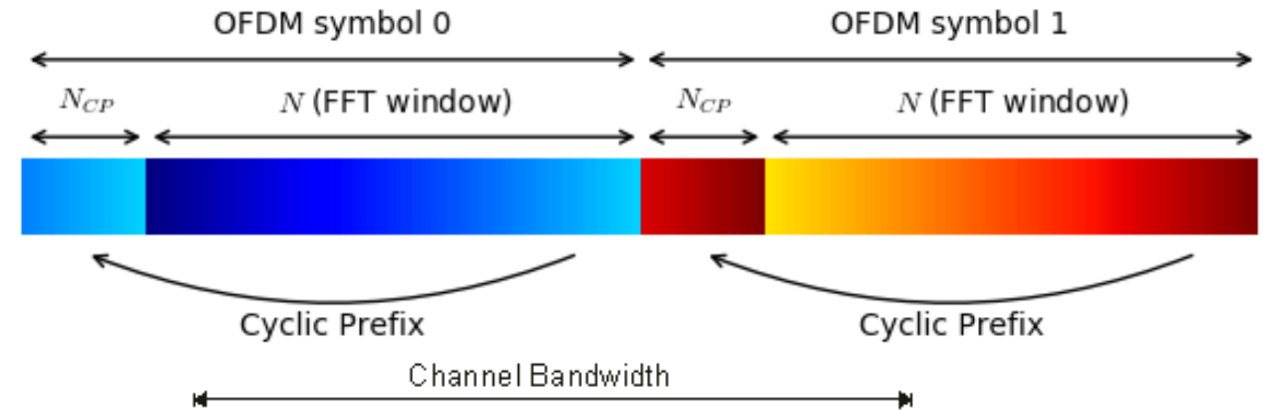
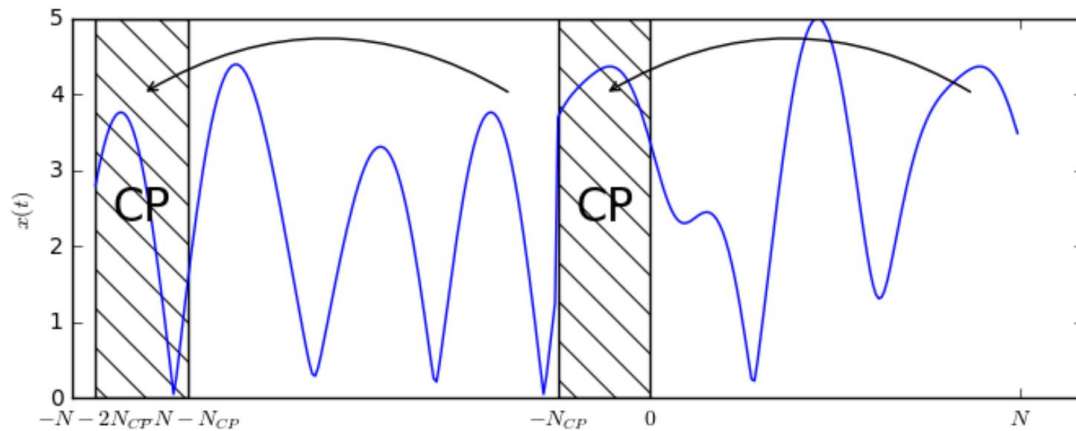
Error protection: The binary stream is protected by :

- Convolutional coding
- Additional Reed Solomon
- Time interleaving



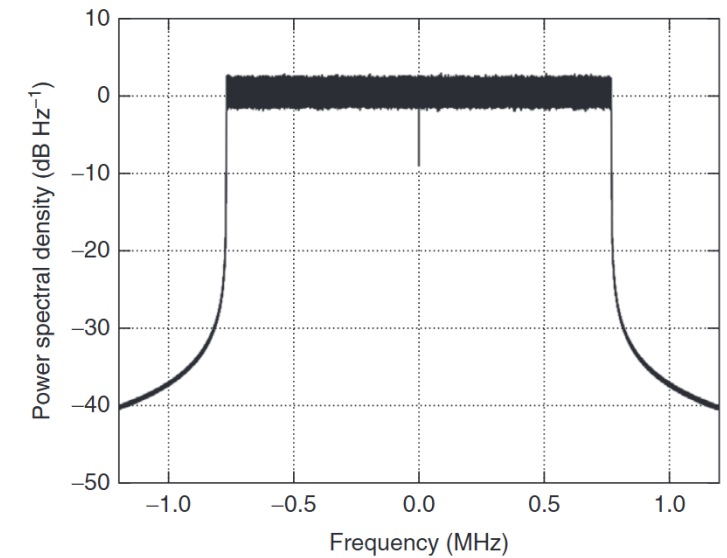
DAB and OFDM:

- Multipath issue -> **ISI (Inter Symbol Interference)** -> **Guard interval**
- Multiple transmitters radiate identical signals on the same frequency -> **Using SFN (Single Frequency Networks)**
 - Each path in an SFN causes only a simple phase shift per subcarrier.
 - The induced phase shift varies from one subcarrier to another, but not within a subcarrier.
 - With the presence of the **CP (Cyclic Prefix)** handling the time spread, there is no spectral overlap : turns the linear convolution with the channel into a circular convolution



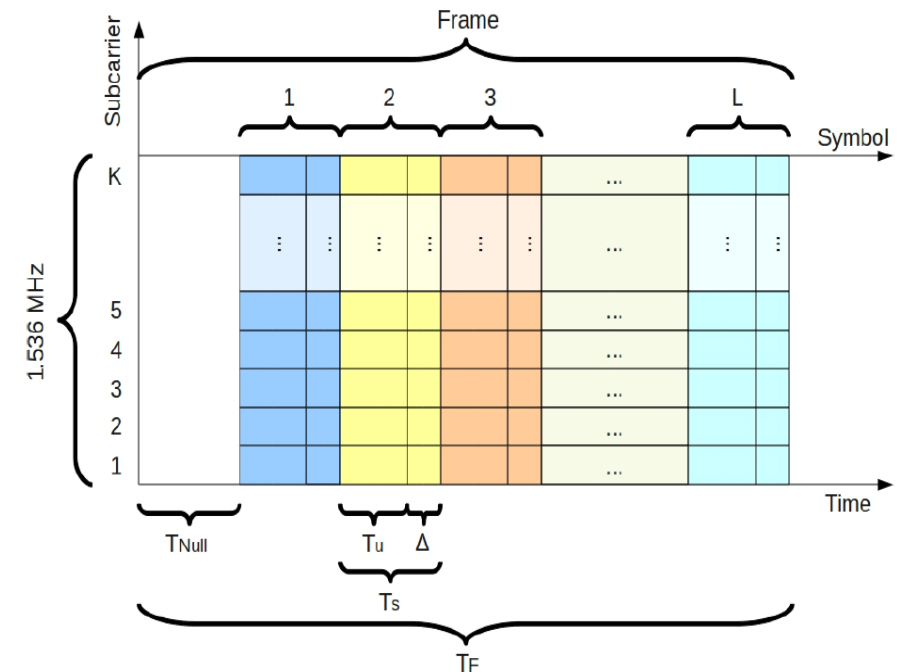
- 1 modulated subcarrier = 1 point in frequency and time
- IFFT creates OFDM Waveform from OFDM Subcarriers
- 1 OFDM symbol = IFFT OFDM Waveform + Guard Interval
- 1 OFDM Burst = one or more OFDM Symbols

$T_f = 196608/F_s$; % transmission frame duration
 $T_{null} = 2656/F_s$; % null symbol duration
 $T_u = 2048/F_s$; % symbol duration excluding ICI (Intercarrier Interference)
 $T_g = 504/F_s$; % duration of ICI
 $T_{sym} = 2552/F_s$; % duration ($T_u + T_g$) of OFDM symbols of indices $l = 1, 2, 3, \dots, L$
 $F_u = F_s/2048$; % Inter-carrier spacing $1/T_u$
 $F_s = 2048000$; % sampling rate of IQ signal
 $K = 1536$; % number of transmitted carriers
 $L = 76$; % number of OFDM symbols per transmission frame (the Null symbol being excluded);



Encoding:

- Each active **subcarrier** carries one complex symbol from differential **QPSK**
- **1536 complex symbols** as spectral data
- **Frequency Interleaver** : prevents localized frequency interference from causing large contiguous data losses by spreading errors across different subcarriers.
- **Mapping** to frequency vector of **2048 samples** :
256 zeros - 768 data - 1 zero (DC) - 768 data - 255 zeros
- **Operate IFFT** -> time domain to what supposed to be harmonic info.
- **With CP** : vector of 2552 with 504 last values of time before all time data
- **First symbol** used as phase reference of differential-QPSK (only 75 symbols are data)



$$y_{l,k} = q_{l,n} \text{ for } l = 2, 3, 4, \dots, L$$

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$$n \in \{0, 1, 2, \dots, 1535\}$$

$$k \in \{-768, -767, -766, \dots, 768\} \setminus \{0\},$$

Table 25: Frequency interleaving for transmission mode I

i	$\Pi(i)$	d_n	n	k
0	0			
1	511	511	0	-513
2	1 010	1 010	1	-14
3	1 353	1 353	2	329
4	1 716	1 716	3	692
5	291	291	4	-733
6	198			
7	1 037	1 037	5	13
8	1 704	1 704	6	680
9	135			
10	218			
11	1 297	1 297	7	273
12	988	988	8	-36
13	1 076	1 067	9	43
14	46			
15	1 109	1 109	10	85
16	592	592	11	-432
17	15			
18	706	706	12	-318
:	:	:		
2044	1 676	1 676	1 533	652
2045	1 819			
2046	1 630	1 630	1 534	606
2047	1 221	1 221	1 535	197

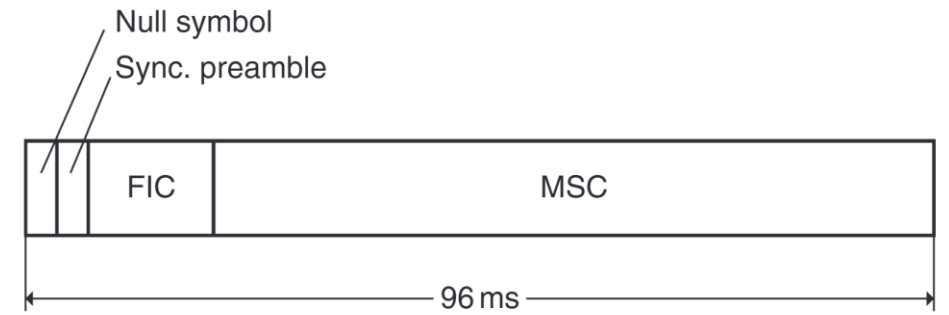
LAB : may 4th

- Compléter les deux fonctions "SymbolFFT.m" et "myFreqDeInterleave.m" du code fournis pour extraire les données du FIC
- Documenter, expliquer et justifier chacune des étapes

```
CRC OK
FrameNr: 2 FIB: 1 Type: 0 Extension: 0 C/N: 0 OE: 0 P/D: 0 EId: 52757 ChangeFlag: 0 ALFlag: 0 CIFCount: 1323 OccurenceChange: 189
FrameNr: 2 FIB: 1 Type: 0 Extension: 2 C/N: 0 OE: 0 P/D: 0 SId: 49701 MSC stream audio ASCTy: 0 SubChId: 5 P/S: 1 CAFlag: 0
                                         SId: 49701 MSC stream audio ASCTy: 0 SubChId: 8 P/S: 0 CAFlag: 0
                                         SId: 49720 MSC stream audio ASCTy: 0 SubChId: 9 P/S: 1 CAFlag: 0
                                         SId: 49704 MSC stream audio ASCTy: 0 SubChId: 8 P/S: 1 CAFlag: 0
FrameNr: 2 FIB: 1 Type: 0 Extension: 1 C/N: 0 OE: 0 P/D: 0 SubChID: 1 StartAdr: 0 UEP ProtLevel: 3 SubChSize: 96 BitRate: 128
FrameNr: 2 FIB: 4 Type: 0 Extension: 1 C/N: 0 OE: 0 P/D: 0 SubChID: 2 StartAdr: 96 UEP ProtLevel: 3 SubChSize: 96 BitRate: 128
                                         SubChID: 3 StartAdr: 192 UEP ProtLevel: 3 SubChSize: 116 BitRate: 160
                                         SubChID: 62 StartAdr: 846 EEP ProtLevel: 3-A SubChSize: 12
                                         SubChID: 4 StartAdr: 308 UEP ProtLevel: 3 SubChSize: 96 BitRate: 128
                                         SubChID: 5 StartAdr: 404 UEP ProtLevel: 3 SubChSize: 48 BitRate: 64
                                         SubChID: 8 StartAdr: 500 UEP ProtLevel: 3 SubChSize: 48 BitRate: 64
                                         SubChID: 11 StartAdr: 692 UEP ProtLevel: 3 SubChSize: 96 BitRate: 128
                                         SubChID: 12 StartAdr: 788 UEP ProtLevel: 3 SubChSize: 58 BitRate: 80
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Synchronization: NULL symbol(1,297 ms) + Synchronization Preamble

- Zero power detection
- FIC / MSC: Different SP (also used as reference of D-QPSK)



Forward Error Correction (FEC): Encoding MSC and FIC

- Convolutional code
 - Code puncturing
 - Time interleave (only for MSC not for FIC) : adjacent error bits issue
- The data is written in rows(solid lines) and read out in columns (dashed lines).
- Reed Solomon code & interleaver (DAB+)

